MINISTRY OF HEALTH · REPUBLIC OF KENYA

Digital Health Agency

Digital Health System Certification Programme

FIELD TEST PREPARATION GUIDE

Read this guide carefully, then submit your endpoints through the DHA portal immediately

APPLICANT / ORGANISATION
Nexora Consulting Ltd

SYSTEM NAME
Vitora HMIS

APPLICATION NUMBER
APP-2026-ETXL5Z

FIELD TEST SESSION ID
To be assigned by Testing Lab

SCHEDULED TEST DATE
To Be Confirmed

TOTAL TEST CASES (YOUR APPLICATION)
23 Test Cases Across 4 Domains

⚠ CONFIDENTIAL — FOR APPLICANT USE ONLY

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Section 1 — Understanding the DHA Field Test

Congratulations. Your application has been approved for field testing — a critical milestone in the DHA Digital Health System Certification Programme. This document was prepared specifically for your application. It lists only the test cases that apply to the features you attested, explains what the DHA testing lab will evaluate, and tells you precisely how to prepare.

⚡ **Immediate action required:** You must submit your API endpoints through the DHA portal as soon as possible. Testing cannot begin until your endpoints are received and verified by the lab team.

What is the Field Test?

The field test is a hands-on technical evaluation conducted by DHA-certified lab staff. The lab team evaluates whether your system behaves exactly as described in your attestation. This is NOT a theoretical review — your system will be tested directly and outcomes observed in real time. There are **23 test cases** scheduled for your application.

Testing Mode: Some tests may be conducted **virtually (online)** where the lab team accesses your system remotely, while others may be conducted **on-site** at your premises or at a DHA facility. Your assigned testing mode is shown on the cover page. Ensure your designated team is available and prepared for the entire testing session regardless of the mode.

The Four Testing Domains

Domain 1 — Functionality

Tests the core clinical features of your EHR/HIS: patient demographics, order entry, prescribing, clinical decision support, and quality measures. The lab team uses your system's UI directly.

Domain 2 — Reporting

Tests disease surveillance and health reporting: immediate disease alerts, IDSR weekly reports, public health event detection, IHR compliance, and routine reporting. Some tests require demonstrating MoH/DHIS2 connectivity.

Domain 3 — Security

Covers data protection compliance, authentication, audit trails, encryption, backup, and data integrity. Split across three methods: document review, live system testing, and technical inspection. See Section 2 for the full security testing explanation.

Domain 4 — Interoperability

Tests data exchange with external systems: Kenya HIE (FHIR), FHIR/SDMX standards compliance, semantic interoperability, and standard terminology usage (ICD-11, SNOMED CT, LOINC). API endpoints must be submitted via the DHA portal.

Testing Methods

Each test case uses one of the following methods. Your preparation must match the method:

Method	What Happens	What You Must Provide	Applies To
🖥 Live System Demonstration	Your technical team demonstrates the system to the lab staff — creating records, placing orders, testing login/logout, entering invalid data, etc. This may be done on-site or remotely (virtually) depending on your assigned testing mode.	System URL, pre-loaded test data, your technical team available to demonstrate	Functionality, most Security tests, Terminology
📄 Document Review	Lab staff reviews compliance documents you provide. No technical system access required for these tests.	Original/certified documents (ODPC certificate, DPIA, DR plan, backup logs). Upload documents via the DHA portal or bring to on-site session.	SEC-DP-001, SEC-BCK-001
🔍 Technical Inspection	Lab staff inspects encryption evidence: opens browser dev tools to confirm TLS/SSL, reviews database encryption config documents, validates certificates.	SSL certificate visible in browser, DB encryption documentation, key management docs	SEC-ENC-001
🔌 API / Integration Test	Lab staff calls your system's API endpoints directly to verify data exchange capabilities. GET endpoints are called and responses validated against your documented schema.	Submit your GET API endpoints via the DHA portal immediately after receiving this email. Include sample responses.	INTER-HIE-001, INTER-STD-001, INTER-SEM-001

Test Scoring

Each test case is scored: **Pass** or **Fail**. All pass criteria listed per test case must be met. Critical tests carry higher weight. Your overall score determines whether DHA proceeds to certificate issuance or requires remediation.

> ⚠ **Important:** Test cases are generated from your attestation. The lab team will cross-reference what you claimed with what the system actually demonstrates. Any feature you attested as implemented must work during testing. If it does not, it counts as a failed test case.

Section 2 — Security Testing & How to Prepare

Security testing is the domain applicants find most confusing. This section explains every security test — what the lab team does, what method they use, and exactly what you must prepare. Security tests are **not penetration testing** — DHA lab staff do not attempt to hack your system. They verify that the security controls you attested are demonstrably in place.

Document Review
You provide compliance documents via the DHA portal or at the test session. Lab staff reads and verifies them.

Live System Demonstration
Your team demonstrates security features to the lab staff hands-on — login, MFA, roles, audit logs. Done on-site or virtually.

Technical Inspection
Lab staff inspects encryption evidence: browser TLS check, DB encryption config, certificate validity.

Test ID	Test Name	Method	How Your Team Demonstrates	What You Must Prepare
SEC-DP-001	Data Protection Compliance	**Document Review**	Reads your ODPC certificate, checks your registration number matches attestation, reviews your DPIA and privacy policy.	ODPC certificate (original/certified copy), completed DPIA document, privacy policy document or accessible URL, data subject rights procedure.
SEC-AUTH-001	User Authentication & Authorization	**Live System Test**	Your team demonstrates: logging in with valid credentials, attempting invalid credentials (should fail after N attempts), enabling and testing MFA, showing different user roles and verifying each can only access permitted screens, demonstrating auto-logoff timeout, and showing break-glass emergency access.	Be ready to demonstrate all authentication and authorization features. Document your role-permission matrix. Confirm auto-logoff timeout is active. Prepare the break-glass access procedure document.
SEC-AUDIT-001	Audit Logging & Trail	**Live System Test**	Your team demonstrates CRUD operations (create, read, update, delete) on patient records, then shows the audit log to verify every action was recorded with user ID, timestamp, and before/after values. Team attempts to edit or delete a log entry (must fail). Lab staff checks log retention period and searches/filters logs.	Ensure audit log viewer is accessible. Verify in advance that CRUD operations are captured. Attempt to manually delete a log entry before the test to confirm it fails — if it succeeds, fix this immediately. Have retention policy documentation.
SEC-ENC-001	Data Encryption	**Technical Inspection**	Your team opens the system URL in a browser → clicks the padlock icon → shows the SSL certificate (issuer, validity, TLS version) to the lab staff. Presents database encryption config document. Shows key management procedures.	Your system must run on HTTPS with a valid SSL certificate (not self-signed). Prepare a one-page document showing: database encryption method (AES-256 recommended), TLS version (1.2 minimum, 1.3 preferred), certificate details, encryption key management procedure,

Test ID	Test Name	Method	How Your Team Demonstrates	What You Must Prepare
			Demonstrates backup encryption.	and evidence that backup files are encrypted.
SEC-BCK-001	Backup & Disaster Recovery	**Document Review**	Reads your backup schedule document, confirms backups are stored off-site, reviews your Disaster Recovery (DR) Plan, checks the most recent DR test results, verifies RTO and RPO targets.	Backup schedule document, off-site backup storage confirmation, DR Plan document, most recent DR test results, RTO/RPO statement. All documents should be dated within the last 12 months.
SEC-INT-001	Data Validation & Integrity	**Live System Test**	Your team demonstrates entering invalid data (future date of birth, negative vital sign values, malformed National ID). All should be rejected. Demonstrates file upload integrity. Shows data consistency across related records.	Pre-test your system: try entering a DOB of year 2099, a weight of -5, a National ID with letters. All must be rejected with a clear error message. Confirm checksum validation is active on file uploads. Ensure referential integrity is enforced (no patient record can be deleted if it has active encounters).

Preparing Your System for Testing

Your production or testing environment must be available for the DHA lab team during the scheduled test window. Ensure the following are in place:

1. Your system must be **running and accessible** at the time of testing. If testing is virtual, the system URL must be reachable from the internet.
2. Use the **same application version** as what you plan to deploy in production. Do not use an incomplete or in-development version.
3. Load the system with **realistic test data**: patient records, encounters, lab results, prescriptions. Do not use an empty system.
4. Your **technical team must be present** to demonstrate the system to the lab staff. They will walk through each test case, showing functionality live.
5. Ensure your **technical team is available** during the entire test session to address any questions or demonstrate additional features as requested.
6. Do not make system changes between receiving the test schedule and the test date.

⚡ Submit Your API Endpoints — Required Immediately

For interoperability and API-related tests, you must **submit your API endpoints through the DHA portal immediately** after receiving this email. The lab team cannot schedule your API tests until endpoints are received and verified.

1. **Log in to the DHA portal** and navigate to your field test session.
2. **Submit each API endpoint** — provide the full URL and HTTP method (GET only for attestation testing). Include a sample response payload so the lab team knows what to expect.

3. **Ensure endpoints are live** — the lab team will call your endpoints directly during testing. Endpoints that time out or return errors will be marked as failed.

4. **For FHIR tests** — submit your FHIR-compliant REST endpoints. The lab team will validate resource structure (Patient, Encounter, Observation) against FHIR R4 standards. Upload sample FHIR resource bundles so the lab team can pre-validate.

5. **Test your endpoints yourself first** — call each endpoint from outside your network to confirm it responds correctly. Fix any issues before submitting.

⚠ **Do not delay endpoint submission.** Your field test timeline depends on it. The lab team will begin reviewing and testing your endpoints as soon as they are received. Late submissions may result in your test being rescheduled.

Need help? If you are unsure which endpoints to submit or how to prepare your system, contact the DHA Certification Unit at **certification@dha.go.ke** immediately. Do not wait until the test date.

API Endpoint Requirements — GET Endpoints Only

> **Critical:** All API endpoints submitted for attestation testing **must be GET endpoints** that return a data payload. POST, PUT, PATCH, and DELETE endpoints **cannot** be used for attestation testing. This section explains why and shows you exactly what a valid endpoint submission looks like.

Why GET, Not POST?

Attestation testing requires the DHA lab team to **retrieve and validate** the data your system returns. The HTTP method determines whether this is possible:

HTTP Method	What It Does	Attestation Use?
GET	Calls your endpoint → your system returns a data payload → the lab team inspects and validates the structure, format, and content of the response.	✅ Required
POST	The lab team sends a data payload *to* your endpoint → your system processes it. The lab team cannot verify what your system returns in a consistent, repeatable way.	❌ Not accepted
PUT / PATCH / DELETE	Modifies or deletes data on your server. These are write operations — they do not return the data payloads the lab team needs to validate.	❌ Not accepted

What Your GET Endpoint Must Do

Every GET endpoint you submit must meet **all** of the following requirements:

Requirement	Detail
HTTP GET method	The endpoint must use the GET HTTP method. Example: `GET https://api.yoursystem.com/v1/records`
Live and reachable	The endpoint must respond within **30 seconds**. No firewall blocks, no timeouts. Test it from outside your network before submitting.
Returns JSON (or XML)	The response body must be structured data — JSON is preferred. Example: `{ "id": "001", "status": "active" }`
Matches your documented schema	The fields, data types, and structure of the response must match your API documentation. Do not submit undocumented endpoints.
Sample response provided	When you submit the endpoint URL, also provide a sample response payload so the lab team can cross-reference during testing.

Example: Valid GET Endpoint Response

Below is an example of what a valid GET response looks like. Your actual fields will differ — this is a structural reference only.

```
GET https://api.yoursystem.com/v1/records/REC-00123

Response (HTTP 200 OK):
{
  "id": "REC-00123",
  "status": "active",
  "name": "Acme Corporation",
```

```
  "created_at": "2026-01-15T08:30:00Z",
  "data_type": "vendor_submission",
  "version": "2.1",
  "attributes": {
    "region": "East Africa",
    "category": "Tier 1"
  }
}
```

What the Lab Team Validates

When testing your GET endpoints, the DHA lab team checks the following:

1. **Field presence** — all fields documented in your API specification are present in the actual response.
2. **Data types** — strings, numbers, booleans, and dates match the types declared in your schema (e.g., `id` is a string, not a number).
3. **Format compliance** — dates are in ISO 8601 format, identifiers match expected patterns, enumerations use valid values.
4. **Response codes** — HTTP 200 for success, appropriate error codes (400, 401, 404, 500) for failure scenarios.
5. **Response time** — the endpoint responds within 30 seconds. Timeouts are treated as failures.

Troubleshooting: Endpoint Not Responding?

If your endpoint times out or returns an error, check the following before resubmitting:

Possible Cause	How to Verify / Fix
Firewall blocking external requests	Test the endpoint from outside your internal network (e.g., mobile hotspot, external VPN).
Server not running	Confirm the server process is running and the port is open: `netstat -an | grep :443`
Wrong URL submitted	Double-check for typos, missing path segments, or incorrect base domain.
Authentication blocking	If the endpoint requires an API key or token, provide them alongside the URL.
SSL/TLS certificate error	Verify your certificate is valid and trusted. Expired certificates cause connection failures.

⚠ **Quick test before submitting:** Run the following from a terminal *outside your internal network* to verify the endpoint is reachable:

```
curl -i -X GET "https://api.yoursystem.com/v1/records" -H "Authorization: Bearer YOUR_TOKEN"
```

Expected: HTTP 200 with a JSON body. If you see a timeout or connection error, the issue is network-level — resolve it before submitting the endpoint.

 Do not submit POST endpoints. If your system only exposes POST endpoints for the data the lab team needs to validate, please contact DHA at **certification@dha.go.ke** to discuss a compatible testing approach before your scheduled test date.

Section 3 — Your Test Cases (23 Total)

The following table summarises all test cases scheduled for your application. Test cases are derived entirely from your attestation — only features you claimed are tested. Detailed preparation checklists and test steps follow in Section 4.

#	Test Case ID	Test Name	Domain	Priority	Method
1	FUNC-DEMO-001	Patient Demographics Capture	Functionality	CRITICAL	Live System Test
2	FUNC-CPOE-001	Computerized Provider Order Entry	Functionality	CRITICAL	Live System Test
3	FUNC-PROB-001	Problem List Management (Kenya National Health Terminology Service)	Functionality	HIGH	Live System Test
4	FUNC-MED-001	Medication Management & HPT Registry	Functionality	HIGH	Live System Test
5	FUNC-CDS-001	Clinical Decision Support System	Functionality	HIGH	Live System Test
6	FUNC-SUMM-001	Clinical Summary Generation	Functionality	HIGH	Live System Test
7	FUNC-EPRES-001	Electronic Prescribing	Functionality	HIGH	Live System Test
8	FUNC-QUAL-001	Clinical Quality Measures	Functionality	MEDIUM	Live System Test
9	REP-IMM-001	Immediate Disease Reporting	Reporting	CRITICAL	Live System Test
10	REP-IDSR-001	IDSR Weekly Reports	Reporting	CRITICAL	Live System Test
11	REP-PHE-001	Public Health Event Detection & Alerting	Reporting	HIGH	Live System Test
12	REP-IHR-001	IHR International Concern Reporting	Reporting	CRITICAL	Live System Test
13	REP-ROUT-001	Routine Health Reporting	Reporting	MEDIUM	Live System Test
14	SEC-AUTH-001	User Authentication & Authorization	Security	CRITICAL	Live System Test
15	SEC-AUDIT-001	Audit Logging & Trail	Security	CRITICAL	Live System Test
16	SEC-ENC-001	Data Encryption (At Rest & In Transit)	Security	CRITICAL	Technical Inspect

#	Test Case ID	Test Name	Domain	Priority	Method
17	SEC-BCK-001	Backup and Disaster Recovery	Security	HIGH	□ Document Review
18	SEC-INT-001	Data Validation & Integrity	Security	HIGH	□ Live System Test
19	INTER-HIE-001	Kenya Health Information Exchange	Interoperability	HIGH	□ API / Integration
20	INTER-STD-001	FHIR Standards Compliance	Interoperability	HIGH	□ API / Integration
21	INTER-SEM-001	Semantic Interoperability	Interoperability	HIGH	□ API / Integration
22	INTER-TERM-001	Standard Terminology Usage	Interoperability	HIGH	□ Live System Test

Section 4 — Detailed Test Cases & Preparation Checklists

For each test case: review the preparation checklist, gather required materials, and ensure your system is configured correctly before the test date.

DOMAIN 1

FUNCTIONALITY

8 Test Cases · 2 Critical · 5 High · 1 Medium

Core EHR/HIS functionality testing. The lab team uses your system's user interface directly. Ensure your system is pre-loaded with test data and all attested modules are enabled.

FUNC-DEMO-001

Patient Demographics Capture

CRITICAL Live System Test

Verify system can capture and store standardized patient demographic data

PRE-TEST PREPARATION CHECKLIST

- Ensure your test environment is running the same software version as your production system
- Create at least 5 test patient records with complete demographics
- Ensure National ID and passport validation rules are active in your test environment

TEST STEPS (LAB PROCEDURE)

1. Create a new patient record
2. Verify sex/gender field is available and validates correctly
3. Verify date of birth field accepts valid dates only
4. Verify residence/address fields are present
5. Verify phone number and contact information fields
6. Verify next of kin contact and relationship fields
7. Verify National ID field with validation
8. Verify Passport number field
9. Verify Birth Certificate number field
10. Save the patient record
11. Retrieve the saved record and verify all data is intact

Test Data: Use test patient: John Doe, Male, DOB: 1990-01-15, ID: 12345678

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ All claimed fields are present in the system
- ✓ Data validation works correctly (e.g., valid ID format)
- ✓ Data structure complies with Kenya National Health Terminology Service standards
- ✓ Saved data persists and can be retrieved accurately

FUNC-CPOE-001

Computerized Provider Order Entry

CRITICAL Live System Test

Verify CPOE functionality for medications, laboratory tests, radiology orders, vital signs

PRE-TEST PREPARATION CHECKLIST

- Create a test patient with an active encounter/visit in your test environment
- Enable all CPOE modules you attested (medications, lab, radiology, vitals)
- Configure routing rules so orders reach the correct department queues

TEST STEPS (LAB PROCEDURE)

1. Login as a healthcare provider
2. Open a patient encounter
3. Create a medication order and verify it processes correctly
4. Order a laboratory test and verify order is sent to lab system
5. Order a radiology examination
6. Record vital signs (BP, temp, pulse, etc.)
7. Verify orders generate billing entries
8. Verify all orders are saved and retrievable

Test Data: Test patient with active encounter

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Orders can be created for all attested categories
- ✓ Orders are properly validated
- ✓ Orders route to appropriate departments/systems
- ✓ Order history is maintained

FUNC-PROB-001

Problem List Management (Kenya National Health Terminology Service)

HIGH Live System Test

Verify problem list functionality with Kenya National Health Terminology Service compliance

PRE-TEST PREPARATION CHECKLIST

- Load Kenya National Health Terminology Service (KNHTS)/ICD-11 code set into the test environment
- Create a patient with at least two existing problems for history review
- Confirm active/resolved toggle is functional

TEST STEPS (LAB PROCEDURE)

1. Add a new problem/diagnosis to patient record
2. Verify problem is coded using KENHDD standards
3. Update problem status (active/resolved)
4. View problem history
5. Search problems by code or description

Test Data: Test diagnosis: Malaria (ICD-11 code)

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Problems can be added, updated, and viewed
- ✓ KENHDD coding standards enforced
- ✓ Problem history is maintained
- ✓ Active vs resolved status tracked correctly

FUNC-MED-001

Medication Management & HPT Registry

HIGH Live System Test

Verify medication lists, allergies, and HPT registry integration

PRE-TEST PREPARATION CHECKLIST

- Pre-load HPT registry data or confirm live API connectivity to HPT
- Create a test patient with a documented allergy (e.g., Penicillin)
- Ensure at least 3 active medications exist on the test patient

TEST STEPS (LAB PROCEDURE)

1. View active medication list for patient
2. View medication history
3. Add allergy to patient record
4. Verify HPT (Health Products & Technologies) registry integration
5. Prescribe a new medication
6. Verify allergy checking triggers warning if applicable

Test Data: Test patient with known penicillin allergy

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Active medications are displayed
- ✓ Medication history is accessible
- ✓ Allergies are checked before prescribing
- ✓ HPT registry provides medication data

FUNC-CDS-001

Clinical Decision Support System

HIGH Live System Test

Verify CDS provides evidence-based interventions

PRE-TEST PREPARATION CHECKLIST

- Configure at least one drug-drug interaction alert in the test environment
- Have a test scenario ready that triggers CDS (e.g., prescribe penicillin to allergic patient)
- Prepare abnormal lab values to trigger CDS alerts

TEST STEPS (LAB PROCEDURE)

1. Create test scenario that should trigger CDS alert (e.g., drug interaction)
2. Verify alert/recommendation is displayed
3. Test CDS with different data sources:
4. - Problem list data
5. - Allergy data
6. - Lab results
7. - Vital signs
8. Verify recommendations are evidence-based and actionable

Test Data: Scenarios: drug interaction, abnormal lab value, critical vital sign

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Recommendations are evidence-based
- ✓ Alerts trigger at appropriate times
- ✓ Multiple data sources are utilized
- ✓ Recommendations are clinically relevant

FUNC-SUMM-001

Clinical Summary Generation

HIGH Live System Test

Verify clinical summary generation for Kenya HIE

PRE-TEST PREPARATION CHECKLIST

- Have a test patient with complete medical history (diagnoses, meds, labs)
- Confirm FHIR export feature is enabled and accessible
- Test the export manually before the audit to confirm it works

TEST STEPS (LAB PROCEDURE)

1. Generate clinical summary for test patient
2. Verify summary is human-readable
3. Verify summary includes: biodata, clinical info, medications, prescriptions, care plan
4. Export summary in Kenya HIE compatible format
5. Validate exported format (FHIR/HL7 compliance)

Test Data: Patient with complete medical history

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Summary is readable by humans
- ✓ Format is Kenya HIE compatible
- ✓ All required sections are present
- ✓ Data is accurate and complete

FUNC-EPRES-001

Electronic Prescribing

HIGH | Live System Test

Verify electronic prescription creation and transmission

PRE-TEST PREPARATION CHECKLIST

- Configure at least one test pharmacy to receive prescriptions
- Confirm electronic prescription transmission is live (not mocked) in staging
- Prepare test prescription: Amoxicillin 500mg TID × 7 days

TEST STEPS (LAB PROCEDURE)

1. Create electronic prescription
2. Include medication, dosage, frequency, duration
3. Attach diagnostic test orders if applicable
4. Transmit prescription electronically to pharmacy
5. Verify pharmacy receives prescription
6. Track prescription status

Test Data: Test prescription: Amoxicillin 500mg TID x 7 days

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Prescriptions can be created
- ✓ Transmission to pharmacy works
- ✓ All prescription elements are captured
- ✓ Prescription is legible and complete

FUNC-QUAL-001

Clinical Quality Measures

MEDIUM | Live System Test

Verify quality measure capture, calculation, and reporting

PRE-TEST PREPARATION CHECKLIST

- Confirm quality measure dashboard is visible and has data
- Pre-load sample data for at least one measure (e.g., vaccination coverage)
- Test PDF/Excel export of quality reports before audit

TEST STEPS (LAB PROCEDURE)

1. Identify quality measures supported by system
2. Capture data for quality measure calculation
3. Calculate quality measures
4. Generate quality measure report
5. Export report in standard format

Test Data: Sample quality measure: Vaccination coverage rate

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Data can be captured
- ✓ Calculations are accurate
- ✓ Reports can be exported

DOMAIN 2

REPORTING

5 Test Cases · 3 Critical · 1 High · 1 Medium

Disease surveillance and health reporting tests. Some tests require demonstrating connectivity to MoH/DHIS2 systems. Use your test environment endpoints for these integrations.

REP-IMM-001

Immediate Disease Reporting

CRITICAL Live System Test

Verify real-time reporting of immediately reportable diseases

PRE-TEST PREPARATION CHECKLIST

- Configure MoH notification endpoint in your test environment
- Confirm the system flags cholera/ebola/other Category 1 diseases automatically
- Show system notification logs from a previous test run if available

TEST STEPS (LAB PROCEDURE)

1. Diagnose patient with immediately reportable disease (e.g., Cholera, Ebola)
2. Verify system flags as immediately reportable
3. Verify automatic notification is triggered
4. Check report is sent to MoH in real-time
5. Verify report includes required data elements per MoH guidelines

Test Data: Test case: Cholera diagnosis

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Report sent in real-time (< 24 hours)
- ✓ Complies with MoH reporting guidelines
- ✓ All required data fields included
- ✓ Notification reaches intended recipients

REP-IDSR-001

IDSR Weekly Reports

CRITICAL | Live System Test

Verify automated IDSR weekly reporting

PRE-TEST PREPARATION CHECKLIST

- Ensure IDSR report generation is configured with DHIS2 staging endpoint
- Pre-load one week of IDSR surveillance data in the test environment
- Test weekly report generation before the audit

TEST STEPS (LAB PROCEDURE)

1. Review IDSR diseases captured during week
2. Generate weekly IDSR report
3. Verify report format matches IDSR requirements
4. Submit report to DHIS2/national system
5. Verify submission confirmation

Test Data: Week of surveillance data

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Reports are auto-generated
- ✓ Submitted weekly without manual intervention
- ✓ Report format matches IDSR standards
- ✓ All priority diseases included

REP-PHE-001

Public Health Event Detection & Alerting

HIGH | Live System Test

Verify detection and reporting of public health events

PRE-TEST PREPARATION CHECKLIST

- Configure outbreak detection thresholds (e.g., 5 similar cases in 3 days)
- Be ready to create 5 measles cases during the test to trigger detection
- Confirm alert routing rules (who receives the alert notification)

TEST STEPS (LAB PROCEDURE)

1. Create scenario for disease outbreak (e.g., cluster of similar cases)
2. Verify system detects unusual pattern/event
3. Verify alert is triggered
4. Check alert notification mechanism
5. Verify event report is generated

Test Data: Simulated outbreak: 5 measles cases in 3 days

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ System detects events automatically
- ✓ Alerts are sent to appropriate authorities
- ✓ Detection thresholds are configurable
- ✓ Event reports contain required information

REP-IHR-001

IHR International Concern Reporting

CRITICAL Live System Test

Verify reporting of events of international concern (IHR)

PRE-TEST PREPARATION CHECKLIST

- Prepare documentation of the IHR escalation pathway from your system to WHO
- Configure test IHR report generation
- Verify report contains all WHO IHR required data fields

TEST STEPS (LAB PROCEDURE)

1. Identify event of potential international concern
2. Verify system flags as IHR-reportable
3. Generate IHR report
4. Verify report format complies with IHR regulations
5. Check escalation pathway

Test Data: Test case: Novel coronavirus case

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ IHR compliance verified
- ✓ Report includes all IHR required elements
- ✓ Escalation to WHO pathway exists

REP-ROUT-001

Routine Health Reporting

MEDIUM Live System Test

Verify monthly, quarterly, and annual routine reports

PRE-TEST PREPARATION CHECKLIST

- Pre-load at least 3 months of historical data in the test environment
- Confirm monthly, quarterly, and annual report templates are configured
- Test PDF and Excel export of routine reports before the audit

TEST STEPS (LAB PROCEDURE)

1. Generate monthly health report
2. Generate quarterly report
3. Generate annual report
4. Verify report completeness and accuracy
5. Export reports in required formats

Test Data: Historical data for report period

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Reports contain accurate aggregate data
- ✓ Required indicators are included
- ✓ Reports can be exported (PDF, Excel, etc.)
- ✓ Data validation checks pass

DOMAIN 3

SECURITY

5 Test Cases · 3 Critical · 2 High

Security and compliance tests. Three different methods apply — document review, live system testing, and technical inspection. Read Section 2 carefully before preparing for these.

SEC-AUTH-001

CRITICAL Live System Test

User Authentication & Authorization

Verify authentication mechanisms and access controls

PRE-TEST PREPARATION CHECKLIST

- Ensure MFA is active and configured on the accounts you will demonstrate
- Confirm auto-logoff timeout is set to 15 minutes in your environment
- Prepare the emergency/break-glass access procedure documentation
- Confirm failed login lockout policy is active
- Be ready to demonstrate login, role-based access, and MFA during the session

TEST STEPS (LAB PROCEDURE)

1. Test user login with valid credentials
2. Test login with invalid credentials (should fail)
3. Verify MFA is required and works
4. Test role-based access control (RBAC)
5. Verify users only access permitted functions
6. Test automatic logoff after 30 minutes
7. Test emergency access mechanism (break-glass)
8. Test password complexity requirements

Test Data: Test users with different roles: admin, clinician, clerk

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Unique user identification enforced
- ✓ RBAC controls work correctly
- ✓ MFA cannot be bypassed
- ✓ Failed login attempts are logged
- ✓ Auto-logoff works after timeout

SEC-AUDIT-001

Audit Logging & Trail

CRITICAL Live System Test

Verify comprehensive audit trail implementation

PRE-TEST PREPARATION CHECKLIST

- Audit log viewing must be accessible to the lab team (admin panel or log viewer)
- Confirm logs capture: user ID, action, timestamp, record affected, before/after values
- Verify audit logs cannot be deleted by any user — including admins
- Confirm log retention policy meets your attested period

TEST STEPS (LAB PROCEDURE)

1. Perform various user actions (create, read, update, delete)
2. Review audit logs for recorded actions
3. Verify logs capture: user, action, timestamp, affected data
4. Attempt to modify audit logs (should fail)
5. Verify version history for data changes
6. Check audit log retention period
7. Test audit log search and filtering

Test Data: Series of CRUD operations on patient records

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ User actions are logged
- ✓ Data changes are tracked with before/after values
- ✓ Logs cannot be altered or deleted
- ✓ Version history is maintained
- ✓ Logs retained for 7 years

SEC-ENC-001

Data Encryption (At Rest & In Transit)

CRITICAL Technical Inspect

Verify encryption of data at rest and in transit

PRE-TEST PREPARATION CHECKLIST

- ⚠ BRING TECHNICAL DOCUMENTS — this test is a technical compliance inspection
- SSL/TLS certificate (browser-navigable — lab staff will inspect via browser)
- Database encryption configuration document (screenshot + method: AES-256, TDE, etc.)
- Key management procedure document (who holds keys, rotation schedule)
- Encryption certificate validity dates + issuing authority
- Proof that backup files are encrypted (sample encrypted backup file or config)

TEST STEPS (LAB PROCEDURE)

1. Verify database encryption (data at rest)
2. Check encryption standard and strength
3. Intercept network traffic and verify TLS/SSL (data in transit)
4. Verify encryption certificates are valid
5. Review key management procedures
6. Test secure file storage for uploaded documents
7. Verify backup files are also encrypted

Test Data: Sample patient data transmission

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Database encryption verified
- ✓ TLS 1.2+ used for all transmissions
- ✓ Encryption standard is AES-128 (Fernet) for data at rest, TLS 1.2+ for data in transit
- ✓ Encryption keys are securely managed

SEC-BCK-001

Backup and Disaster Recovery

HIGH Document Review

Verify backup procedures and disaster recovery capabilities

PRE-TEST PREPARATION CHECKLIST

- ⚠ BRING PHYSICAL/DIGITAL DOCUMENTS — this test is a document review
- Backup schedule document (daily/weekly/monthly — what gets backed up)
- Off-site backup storage confirmation (location, provider)
- Disaster Recovery Plan document
- Most recent DR test results/report
- RTO (Recovery Time Objective) and RPO (Recovery Point Objective) statements

TEST STEPS (LAB PROCEDURE)

1. Review backup schedule and procedures
2. Verify backup frequency: Daily
3. Verify off-site backup storage
4. Review disaster recovery plan documentation
5. Review recent DR test results
6. Verify RTO and RPO meet requirements
7. Test backup restoration process (sample data)

Test Data: RTO: 4 hours, RPO: 24 hours

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Backups performed Daily
- ✓ Off-site backups verified
- ✓ DR plan exists and is comprehensive
- ✓ DR plan has been tested recently
- ✓ Backup restoration works correctly

SEC-INT-001

Data Validation & Integrity

HIGH Live System Test

Verify data integrity mechanisms

PRE-TEST PREPARATION CHECKLIST

- Prepare invalid test inputs to demonstrate validation (future DOB, negative age, malformed IDs)
- Confirm database referential integrity is enforced (no orphaned records)
- Ensure checksum or hash validation is visible for file uploads
- Test digital signature functionality (if attested) before the audit

TEST STEPS (LAB PROCEDURE)

1. Test data validation rules (e.g., invalid date should be rejected)
2. Attempt to enter malformed data (should fail validation)
3. Verify checksum validation for file uploads
4. Test digital signatures for critical documents
5. Test data consistency across related records
6. Verify referential integrity in database

Test Data: Invalid inputs: future dates, negative values, malformed IDs

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Validation rules prevent bad data
- ✓ File integrity verified
- ✓ Digital signatures work
- ✓ Data consistency maintained

DOMAIN 4

INTEROPERABILITY

4 Test Cases · 4 High

Data exchange and standards compliance tests. Kenya HIE and FHIR tests require API access. Submit your API endpoints via the DHA portal immediately.

INTER-HIE-001

Kenya Health Information Exchange

HIGH API / Integration

Verify integration with Kenya HIE

PRE-TEST PREPARATION CHECKLIST

- ⚠ API INTEGRATION TEST — submit your API endpoints via the DHA portal immediately
- API must be able to: POST patient to HIE, GET patient from HIE, show transaction logs
- FHIR resource compliance: Patient, Encounter, Observation resources must be valid
- Consent management must be demonstrable
- Ensure your endpoints are live and reachable from external networks

TEST STEPS (LAB PROCEDURE)

1. Send patient data to HIE
2. Receive patient data from HIE
3. Verify data format compliance (FHIR)
4. Test patient matching/identifier resolution
5. Verify data mapping to HIE standards
6. Test consent management for data sharing
7. Verify audit logging of HIE transactions

Test Data: Test patient record with complete demographics

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ HIE connection established
- ✓ Data sent to HIE successfully
- ✓ Data received from HIE successfully
- ✓ R4 compliance verified

INTER-STD-001

FHIR Standards Compliance

HIGH API / Integration

Verify compliance with FHIR data exchange standards

PRE-TEST PREPARATION CHECKLIST

- ⚠ API INTEGRATION TEST — export FHIR resources for lab team review
- Generate sample FHIR R4 Patient and Encounter resources from test data
- Generate FHIR Observation and DiagnosticReport resources
- Validate resources against FHIR R4 profiles before the test — lab staff will re-validate

TEST STEPS (LAB PROCEDURE)

1. Test FHIR R4 resource operations
2. Validate FHIR resource structure against published profiles
3. Verify resource parsing and interpretation
4. Test error handling for malformed FHIR resources

Test Data: Sample FHIR resources (Patient, Observation, DiagnosticReport)

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ FHIR R4 resources valid
- ✓ Resources are well-formed and parseable
- ✓ Standard FHIR codes and terminology used

INTER-SEM-001

Semantic Interoperability

HIGH API / Integration

Verify semantic interoperability capabilities

PRE-TEST PREPARATION CHECKLIST

- Demonstrate bidirectional data exchange with at least one external system
- Show that coded data (ICD-11 diagnoses, LOINC labs) maps correctly on both ends
- Prepare export of clinical data with coded terminology for lab team review

TEST STEPS (LAB PROCEDURE)

1. Exchange structured data with external system
2. Verify data meaning is preserved (semantic level)
3. Test vocabulary mapping (local codes to standard codes)
4. Verify automated data interpretation
5. Test bidirectional data exchange

Test Data: Clinical data with coded diagnoses and procedures

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Structured data exchange works
- ✓ Semantic meaning preserved

INTER-TERM-001

Standard Terminology Usage

HIGH | Live System Test

Verify use of SNOMED CT, ICD-11, LOINC terminology standards

PRE-TEST PREPARATION CHECKLIST

- ICD-11 code lookup must be functional in the diagnosis entry screen
- SNOMED CT code search must be functional
- LOINC code assignment for lab tests must be demonstrable
- Confirm code sets are current versions (not outdated editions)

TEST STEPS (LAB PROCEDURE)

1. Test ICD-11 coding for diagnoses
2. Test SNOMED CT for clinical terms
3. Test LOINC for lab tests
4. Verify code validity against standard code sets
5. Test code search and lookup functionality
6. Verify code descriptions are correct

Test Data: Sample diagnoses, procedures, lab tests

PASS CRITERIA — ALL CRITERIA MUST BE MET TO PASS THIS TEST

- ✓ Valid ICD-11 codes used
- ✓ SNOMED CT codes valid
- ✓ LOINC codes for labs
- ✓ Code sets are current versions
- ✓ Codes map correctly to descriptions